

Programme	B.Sc Computer Science				
Course Title	Mastering Content Management Systems				
Type of Course	Minor				
Semester	III				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Familiarity with web content management systems (CMS). 2. Basic knowledge of internet technologies provides a foundation for learning web design.				
Course Summary	The course covers fundamental web design concepts, including HTML and CMS principles, and focuses on Drupal as a robust content management system. Students will learn to create and customise websites using Drupal, exploring its features, such as content types, themes, and modules to build dynamic and interactive web pages.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Cultivate a robust understanding of web design fundamentals, laying a strong foundation for their journey into the dynamic world of digital design and development.	U	C	Assignment / Instructor-created exams / Quiz
CO2	Attain comprehensive knowledge and practical proficiency in Content Management Systems (CMS), empowering to navigate and excel in the ever-evolving landscape of digital content creation and management.	U	C	Assignment / Instructor-created exams / Quiz

CO3	Develop expertise in Drupal, a widely used CMS platform, gaining comprehensive understanding of its features, configuration, and installation processes, thus preparing them for proficient and innovative web development endeavors.	Ap	P	Practical Assignment / Instructor-created exams / Quiz
CO4	Impart a comprehensive understanding of website development using Drupal and facilitate the acquisition of expertise across various options within the Drupal ecosystem.	Ap	P	Practical Assignment / Instructor-created exams / Quiz
CO5	Gain an understanding of how to apply web design concepts to real-world scenarios, effectively designing and developing functional and aesthetically pleasing websites utilizing the Drupal CMS.	C	P	Practical Assignment / Instructor-created exams / Quiz
CO6	Develop proficiency in advanced website management skills, including installing and configuring modules, managing menus, and more, to effectively navigate and optimize the functionality of websites built on the Drupal platform.	C	P	Practical Assignment / Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I		Introduction to Web Designing	08	15
	1	Basics of Web Designing -World Wide Web (WWW), W3C, Web Browser	2	
	2	Web Server, Web Hosting, Web Pages	2	
	3	Domain Name System, Uniform Resource Locator	1	
	4	Overview of HTML: Definition and Basic structure	3	

II	Introduction to CMS		09	15
	5	Introduction to Content Management Systems (CMS) - Features of CMS	2	
	6	Web Content Management System	2	
	7	Components of Content Management System	2	
	8	Enterprise Content Management System	3	
III	Introduction to Drupal		13	20
	9	Drupal - Features, Advantages and Disadvantages,	2	
	10	Comparison of Wordpress and Drupal	1	
	11	Installation and Configuration	2	
	12	Content types and Field	2	
	13	Drupal Architecture	2	
	14	User Management, Managing Comments	2	
	15	Creating and Customizing Themes	2	
IV	Building Website		15	20
	16	Website Development - Working with Templates and Template files	2	
	17	Articles, Creating Web Forms	2	
	18	Managing blocks, Add Links to Blocks, Moving Elements within Block	3	
	19	Blocks and Regions	2	
	20	Creating and Customizing Views	2	
	21	Installing and Configuring Modules	2	
	22	Static Pages, Creating Pages, Menu Management.	2	
V	Hands-on Programming		30	30
		1. Install Drupal on your local server and configure it to run. 2. Create a new content type called "Blog Post" with fields for title, body, and image. 3. Add a new field to the user profile for "Job Title" using Drupal's field management system. 4. Customise the default theme by changing the colours and		

		fonts. 5. Create a new custom theme from scratch and apply it to your Drupal site. 6. Add a new block to the sidebar displaying recent blog posts. 7. Create a custom view that displays a list of all users with their job titles. 8. Configure Drupal's built-in caching system to improve performance. 9. Install and configure a contributed module from Drupal.org to extend the functionality of your site. 10. Implement a custom module that adds a new feature to your Drupal site, such as a contact form or slideshow. 11. Set up user permissions to restrict access to certain parts of the site based on user roles. 12. Use the Drupal Views module to create a dynamic page that displays a grid of images from a specific content type. 13. Implement a responsive design for your Drupal site using CSS media queries. 14. Use Drupal's taxonomy system to categorise content and create a navigation menu based on taxonomy terms. 15. Test your site's accessibility using automated tools and make any necessary adjustments to improve accessibility for users with disabilities.		
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Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	3	1	1	3	1						
CO 2	1	3	2	1	3	1						
CO 3	1	3	1	1	3	2						
CO 4	1	3	3	1	3	2						
CO 5	3	3	3	1	3	2						

CO 6	1	3	3	1	3	2						
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Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓

References:

1. Jennifer Campbell, Jennifer T Campbell, Web Design: Introductory, Course Technology.
2. Jason Beaird and Alex Walker, The Principles of Beautiful Web Design, SitePoint.
3. Bob Boiko, Content Management Bible, Wiley.
4. Daniel Sipos, Drupal 9 Module Development, Packt Publishing Limited